TASK (1)

Conceptual design using ER model :- 28/7/25

Aim:- Conceptual Design using ER model for college Management system using draw.io

Tools Required:- https://draw.io (or creatly / ERD PIUS)

Steps involved in creating ER diagram.

Step 1:- understanding

* Analyze the real world applications: college management system.

* understand domain : student, addmission, Lecture, subjects.

Step 2:- Identify, major Entities

Entities are core components resprosending objects (or) concepts:

- students
- Admission
- Timetable
- Lecture
- Subjects.

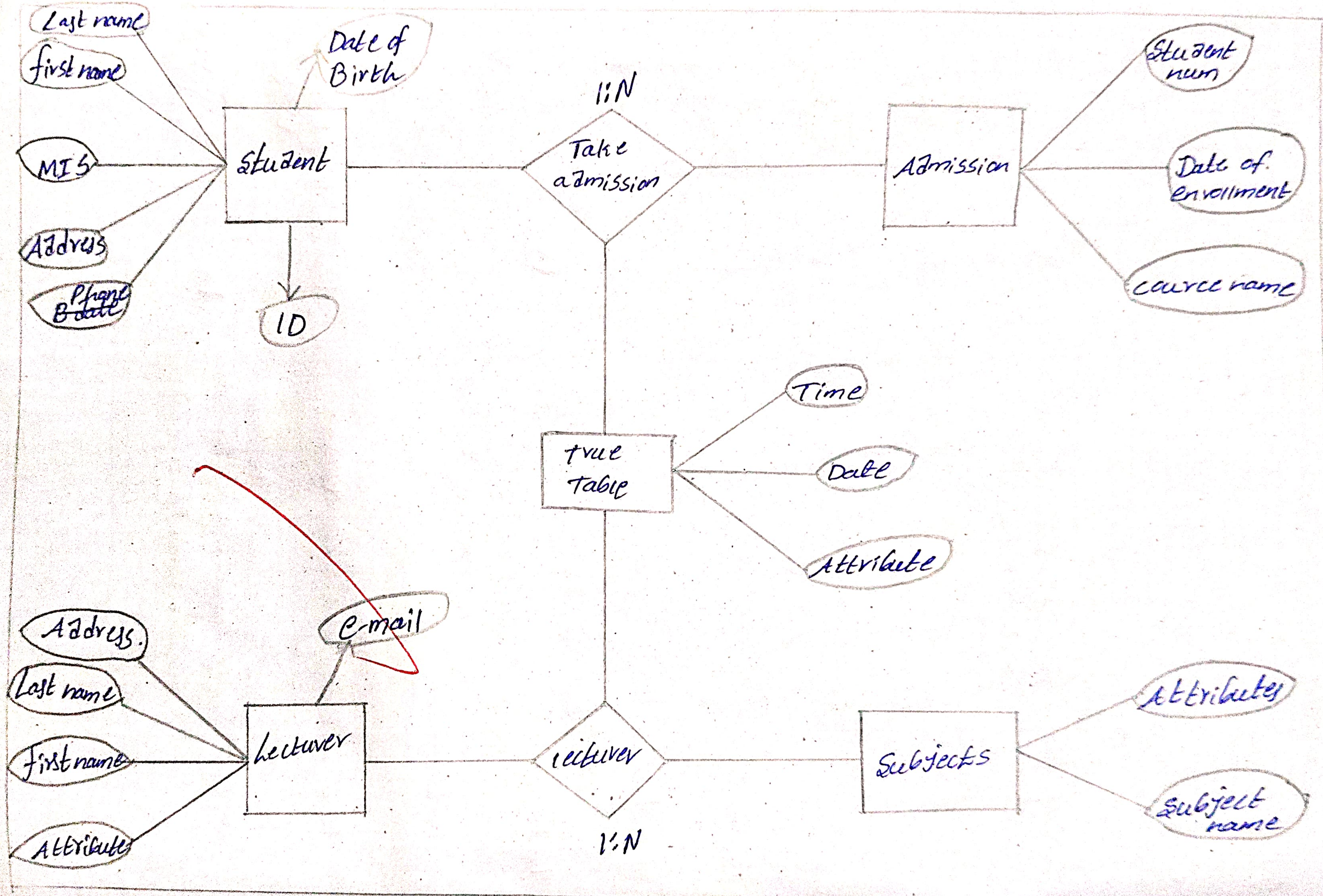
Step 3:- Identity, Attributes for entity.

Entity Attributes:

Student:- Name, student, ID, Address, DOB, Department

Addmission :- Admission-num, course name, date of enrollement, student ID.

Timetable:- Time, date, classes.


Last name
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Student
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Time
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Date
Attribute
e-mail
Address.
Last name
first name
Attribute
Lecturer
lecturer
1:N
Subjects
Attributes
Subject name

Lecturer:- Name, Lecturer. ID, Gender, department, Ph-num.

Subjects:- subject-name, subject code.

Step 4:- Relationship b/w entities.

* students take one (or) more Admissions.

* Admission student gets timetable

* Time table gives one (or) more lecturers.

* Lectures teachers one (or) more subjects.

steps:- Draw ER diagram using draw.io

* open https://draw.io

* choose blank diagram → click create

* from left panel, draws the following

* use ellipes for attributes.

* connect using lines:

* solid lines for relationship connectors.

Input for ER design

Real_time college management system

Scenario - use requirements.

Data base design rules Entities - Attribute - Relationship).

Output:-

Entity relationship diagram (ERD) that shows.

* All identified entities with attributes

* All relationship with appropriate cardinalities.

* Foreign keys and keys marked appropriately.

Result:- Conceptual design using ER Model for college management system using draw.io has been Implemented successfully and task is done.

VEL TECH	
EX No.	4
PERFORMANCE (5)	2
RESULT AND ANALYSIS (5)	3
VIVA VOCE (5)	5
RECORD (5)	2
TOTAL (20)	12
SIGN WITH DATE	28/2/5

Result:- This takes helped to understand the importance of concepted design in database management using draw.io; we were able to visually model to a real-time college management system into an ER diagram.

(1.b)

Relational model:-

Student	
Name	VARCHAR
student_ID(PK)	INT
Department_ID	INT
Department	VARCHAR
DoB	DATE
Adress	VARCHAR

Time table	
Time	TIME
Date	DATE
Classes	VARCHAR
TT.ID (PK)	INT

Admission	
Student ID(PK)	INT
Admission-num	INT
course name	VARCHAR
Date of enrollment	DATE

Lecturer	
Name	VARCHAR
Gender	VARCHAR
Lecture-ID(PK)	INT
Ph-num	VARCHAR
Department	VARCHAR

Subjects	
Subject_Name	VARCHAR
Subject code (PK)	INT

TASK (1.6)

CONVERT ER DIAGRAM INTO RELATIONAL MODEL.

Aim:- To convert the ER diagram into relational model.

steps for converting ER diagram to the relational Model

* Entity type become a table
* All single valued attributes become a column for the table.
* A key Attribute of the entity type represented by primary key.
* The multi valued attribute is represented by a separate table.
* Composite attribute represent by components.
* Derived attributes are not considered in the Table.

Using these rules, you can convert the ER Diagram to tables & coloum and assign the Mapping between the tables.

VELTECH	
EX No.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	

Result:- The relational model for the given ER Diagram was successfully converted.